Indr 252 Applied Statistics
Department of Industrial Engineering
Koç University

Fall 2025

Syllabus

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BS&MS in IE (ODTÜ)

PhD in Operations (Case Western Reserve University)

Post-doctorate at EURANDOM in the Netherlands (A research institute on random processes)

At Koç University Industrial Engineering since 2001

Research interests: Health care applications, stochastic models

Personal interests: Books, cats, volunteering, women in STEM



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Plan of the day

Syllabus

Sampling and statistics

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Indr 252 Applied Statistics

Teaching assistants: Ece Ceylan, Arash Gandomkar Ghalhar, Mustafa Furkan Zenginoğlu

Textbook: Statistics for Engineers and Scientists, 5th Edition (ISBN-13: 978-1260547887), W. Navidi, McGraw-Hill Education

McGraw-Hill Education Connect: Textbook companion, https://connect.mheducation.com/connect/container/launch.htm

You need buy the book and access code to be able to accomplish the reading assignments etc.

Software to carry on the statistical analyses: Python

- Num Python: https://numpy.org/doc/stable/reference/routines.statistics.html
- Sci Python: https://docs.scipy.org/doc/scipy/reference/stats.html

All information about the course: Edit course: INDR252-Fall25-01-CE-APPLIED STATISTICS | Learn

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Course Description

- Visual tools to understand the data: all kinds of charts
- How to summarize data: useful statistics such average, standard deviation, etc.
- How to characterize the main properties of processes using data: parameter estimation, confidence intervals, and hypothesis testing.
- How to characterize the underlying probabilistic features using data: Distribution fitting, goodness of fit tests. Independence tests and contingency tables.
- How to relate different variables: Simple linear regression and correlation analysis. Nonlinear and multiple regression, analysis of categorical data.

Industrial engineering applications in quality control and demand forecasting. Statistical software packages and computer implementations.

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Upon completion of this course, you should be able to:

Formulate a problem about a system or process as a statistical question

Design an appropriate data set and collect the data

Present the data in visual forms that help understanding potential problems

Understand and carry on the required statistical analyses to answer the chosen question

Note the shortcomings of the statistical analyses due to the underlying assumptions and data collection process

Assess statistical inferences on the underlying problem.

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Learning experiences

Learning material: introducing the statistical techniques that IEs use

Experiencing:

- Letting the students experience being an IE through cases.
- Labs and case studies are designed so that students actively analyze data using appropriate software.

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Tentative schedule of the course

	Lec 1	Lec 2	Lab
Week 1	Introduction	Random sampling and data description	Lab 1 - Statistical functions in Phyton
Week 2	Random sampling and data description	Propagation of error	Lab 2 - Generating random variables, descriptive stats & plots
Week 3	Propagation of error	Point estimation	Lab 3 - Analyzing Datasets
Week 4	Point estimation	**Cumhuriyet Bayramı**	---
Week 5	CLT/Interval estimators	Probability plots	Lab 4 - CI
Week 6	Interval estimators	Interval estimators	Sample Case
Week 7	Interval estimators	Interval estimators	**Case Exam 1**
Week 8	Interval estimators	Tests of hypotheses	Lab 5 - HT
Week 9	Tests of hypotheses	Tests of hypotheses	PS
Week 10	Tests of hypotheses	Tests of hypotheses	**Midterm**
Week 11	Linear regression	Linear regression	Lab 6 - χ^2 Distribution Tests
Week 12	Linear regression	Linear regression	PS
Week 13	Multiple regression	Multiple regression	Lab 7 - Multiple Regression
Week 14	Multiple regression	Multiple regression	**Case Exam 2**

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Assessment Methods

Method	Description	Weight %
Case Study 1	Case Exam	10.00
Case Study 2	Case Exam	15.00
Midterm	Exam	25.00
Final Exam	Comprehensive Final exam	30.00
Reading Assignments	Assignments from the book	5.00
Participation	Participation (in-class exercises)	5.00
Participation	Lab work (Lab assignments)	10.00
Case Study 1	Case Exam	10.00
Case Study 2	Case Exam	15.00

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Lectures

Lectures: In-class participation: class exercises (5%)

Students will be given straightforward questions in the lectures as class exercises. Students can work on these exercises in groups of two (never more than two). Note that putting in an honest effort to solve the assignment will result in full credit.

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Labs (10%)
Mondays at 17:30-18:40 in SNAB149

- Aim: to teach how to carry out statistical computations using Python and how to interpret the computations and plots.
- You will have access to all related files on the previous Friday. All lab sessions, except for the first one, have a short assignment.
- The students are supposed to work on these assignments before the lab session. There are two main tasks: (1) preparing the Python code that computes all the necessary statistics and generates the plots. (2) interpreting these statistics and plots. The questions in the assignment guide the students to explore the meaning of the statistics and plots. The students are expected to complete the Python code, think about these questions, and start writing their reports before the lab session.

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Labs (10%)
Mondays at 17:30-18:40 in SNAB149

- Students may submit their work without attending the lab session, or they may join the session, where the questions will be discussed at length, and then submit their work. Note that the submission should consist of the Python code, an explanation of how the code is written (AI may be used to write the code, but how it is used should be documented), and the report. The resulting report should answer the questions and show their understanding of the topics covered in the lab.
- During the lab sessions, we will help those who have difficulty with the Python code. However, the main purpose of the sessions is to discuss certain statistical concepts and to help students correctly interpret the computations and plots.
- The lab submission will be due by midnight on the day of the lab session.

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PSs & Sample case

- Sample case: all students should read and think about the case before the session; during the sessions, random groups will work on one part of the case, and then we will discuss the results altogether. You will be given a short test on LearnHub at the end of this session. (as part of lab assessments)
- Problem sessions: problems on the course topics will be solved by the TAs

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Reading assignments (5%):

The corresponding book chapters are assigned for reading before/after the lectures, so that you can be prepared for the new subjects and digest the material covered in class.

Once you read the related chapters, you will be presented a number of questions on the basic concepts. We will hold Q&A sessions in the beginning of each lecture on the latest reading assignment to discuss your questions and comments on these concepts.

These assignments will help you to assess your level of understanding in the course.

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Exams

One midterm (25%): multiple-choice questions in computer labs.

Two case exams (10% & 15%): Group exam; outputs are a Python file and a report

Final exam (25%): A comprehensive exam with multiple-choice questions in computer labs.

Make-up Exam Policies: A make-up exam for the case exam and a make-up exam for other tests will be given to those who have a valid excuse at the end of the semester. Both make-up exams will be comprehensive.

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Case Exams

- On two Mondays through the lecture and lab sessions (16:00-19:00).
  - in groups in the computer labs, where physical presence is mandatory. Students can form their groups. The students without a group will be assigned to certain groups by the teaching team.
  - Students will not have access to cell phones, calculators, or the internet during the exams.
  - In these exams, students are supposed to analyze a case statistically, which will require using Python code. Note that students are not allowed to access any AI tool; instead, some partial code will be shared with the students during the exam. Hence, students need to learn how to write or modify Python code according to their specific needs.

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LearnHub and Connect

I will share the information how to register Connect and access the e-book on Wednesday.

Reading assignments:

- Aim to help you in following the course.
- Some midterm questions will be similar to these exercises.

If you cannot afford buying the book, please contact me.

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Academic Dishonesty

Koç University expects all its students to perform course-related activities in accordance with the rules set forth in the Student Code of Conduct (http://vpaa.ku.edu.tr/academic/student-code-of-conduct). Actions considered as academic dishonesty at Koç University include but are not limited to cheating, plagiarism and collusion. Copying from others or providing answers or information, written or oral, to others is cheating. Copying from another student's paper or from another text without written acknowledgment is plagiarism. Unauthorized help from another person or having someone else write one's paper or assignment is collusion.

Cheating, plagiarism and collusion are serious offenses resulting in an F grade and disciplinary action.

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Other issues

Grades: To be announced in LearnHub. All objections to grades should be made within 7 days of their announcement.

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Why Statistics Matters in Industrial Engineering?

"Without data, you're just another person with an opinion." — W. Edwards Deming

How many of your daily or engineering decisions are truly based on data?



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Everyday Variation

Why Did My Flight Get Delayed?

- Which airline is more reliable?
- Does the average delay tell the full story?

→ Statistics helps us understand variation, not just averages.

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Decision Under Uncertainty

Should We Buy a New Machine?

Current defect rate: 2%

New machine: 1.5%

Sample: 200 parts → 3 defects

Would you invest 1 million TL based on that?

→ Statistics quantifies confidence.

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Statistical Thinking

Statistical inference = separating signal from noise.

This is the backbone of data-driven industrial decisions.

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Amazon's A/B Test

Amazon changes button color → Sales ↑ 3%.

Is it real or random?

Experiments = foundation of Design of Experiments (DOE).

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Industrial Experiments

Examples:

- Testing a new assembly method
- Trying a new supplier material
- Changing oven temperature

Same logic = process optimization.

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The Curious Case of the Broken Bolt

Measurements all within spec, but drifting upward.

Would you stop production?

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Process Stability

Control charts show process health.

Predictable = good.

Statistics = early warning system.

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Recap

Flights → Variation

Machine → Decision under uncertainty

Amazon → Experiments

Bolts → Process monitoring

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What You'll Learn

Descriptive statistics

Probability & sampling

Inference & regression

Design of experiments

Statistical quality control

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Real-World Impact

Manufacturing → Process improvement

Logistics → Forecasting demand

Tech → A/B testing

Industrial engineers = translators between data and decisions.

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Closing Thought

"Statistics is not about numbers — it's about learning from data."

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